

Phase 6: Project Testing

AI Comic Story Creator — SmartBridge Project Submission

1. Testing Levels

- **Unit testing:** layout engine bounding-box calculations for each panel count (1, 2, 3, 4, 5, 6 panels).
- **Schema testing:** validate that Gemini's JSON output always matches the expected script/panel schema.
- **Integration testing:** full pipeline run from a sample premise through to a finished exported PDF.
- **Visual/regression testing:** character visual anchors stay consistent across multiple panels and pages.
- **Load/queue testing:** Cloud Tasks/Celery retry behavior under simulated Imagen rate-limit errors.

2. Sample Test Cases

Test Case	Expected Result
Generate script for a 7-page sci-fi premise	Valid JSON with 7 pages, characters array populated
Render a single panel from a visual anchor prompt	Image returned matching described character traits
Same character across 3 different panels	Character's key visual traits remain recognizably consistent
4-panel grid layout calculation	4 non-overlapping bounding boxes fitting within page margins
Imagen API returns a rate-limit error	Task is retried automatically rather than failing the job
Export finished issue	Print-ready PDF (300 DPI) and/or CBZ file produced without errors

3. Known Limitations to Document

- Prompt-based character consistency is not as reliable as true model fine-tuning; occasional drift is expected.
- Generation cost and latency scale with page/panel count, which should be communicated to end users.